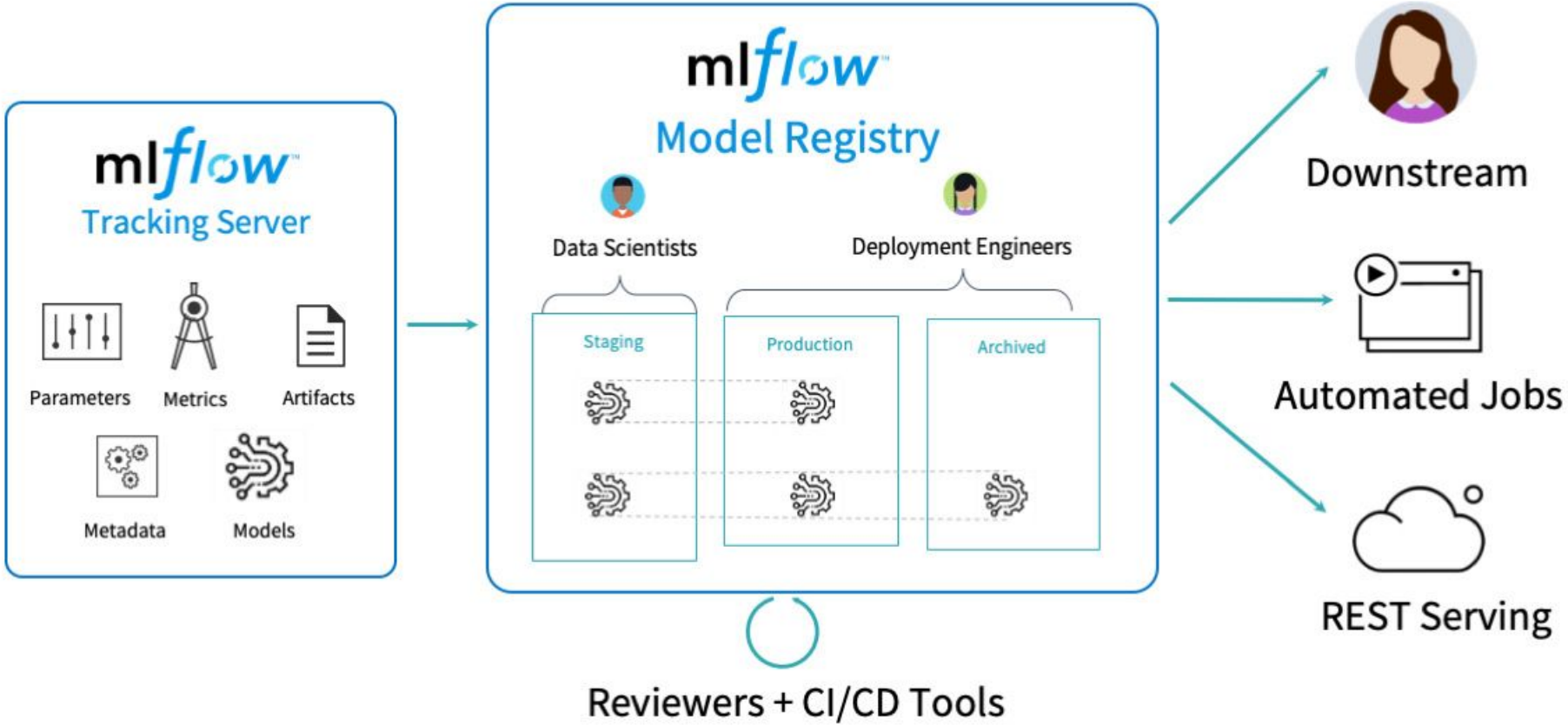
MLFlow

MARSAIL

MLflow for MARS (Motor AI Recognition Solution)

Building a scalable MLOps system for AI-based vehicle recognition

- Company: MARS (Motor AI Recognition Solution)
- Focus: Computer Vision / Vehicle Detection / AI Recognition
- Tool: MLflow


mlflow
Tracking Server
Parameters
Metrics
Artifacts
Metadata
Models
mlflow
Model Registry
Data Scientists
Deployment Engineers
Staging
Production
Archived
Downstream
Automated Jobs
REST Serving
Reviewers + CI/CD Tools

Challenges in AI model development at MARS

- Multiple experiments with YOLO / detection models
- Hard to track which model performs best
- No centralized model version control
- Difficult collaboration between engineers
- Lack of reproducibility in training

Without MLOps

- Models scattered in folders
- Metrics stored in Excel or notebooks
- Hard to reproduce results
- Deployment is manual and error-prone

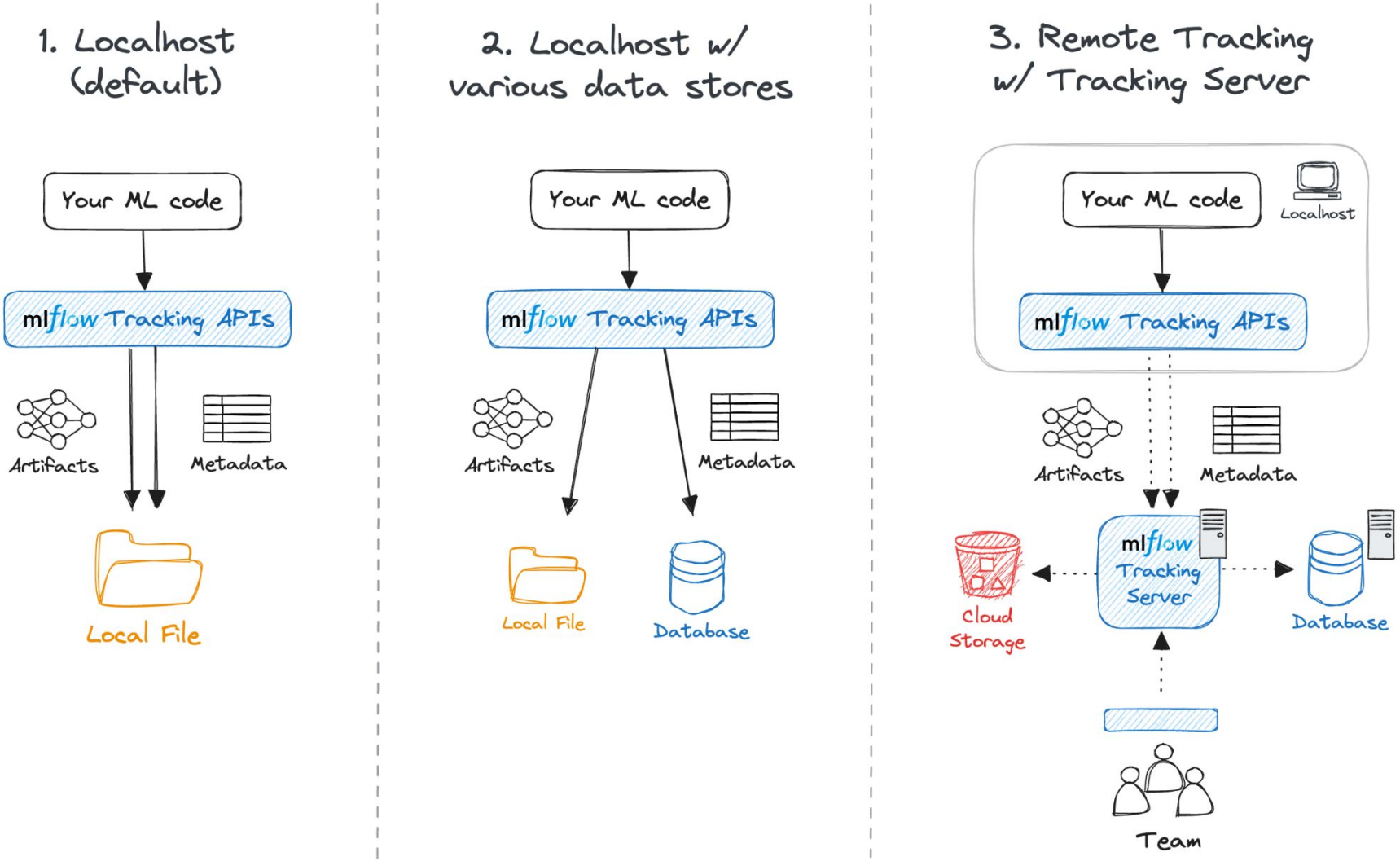
With MLOps

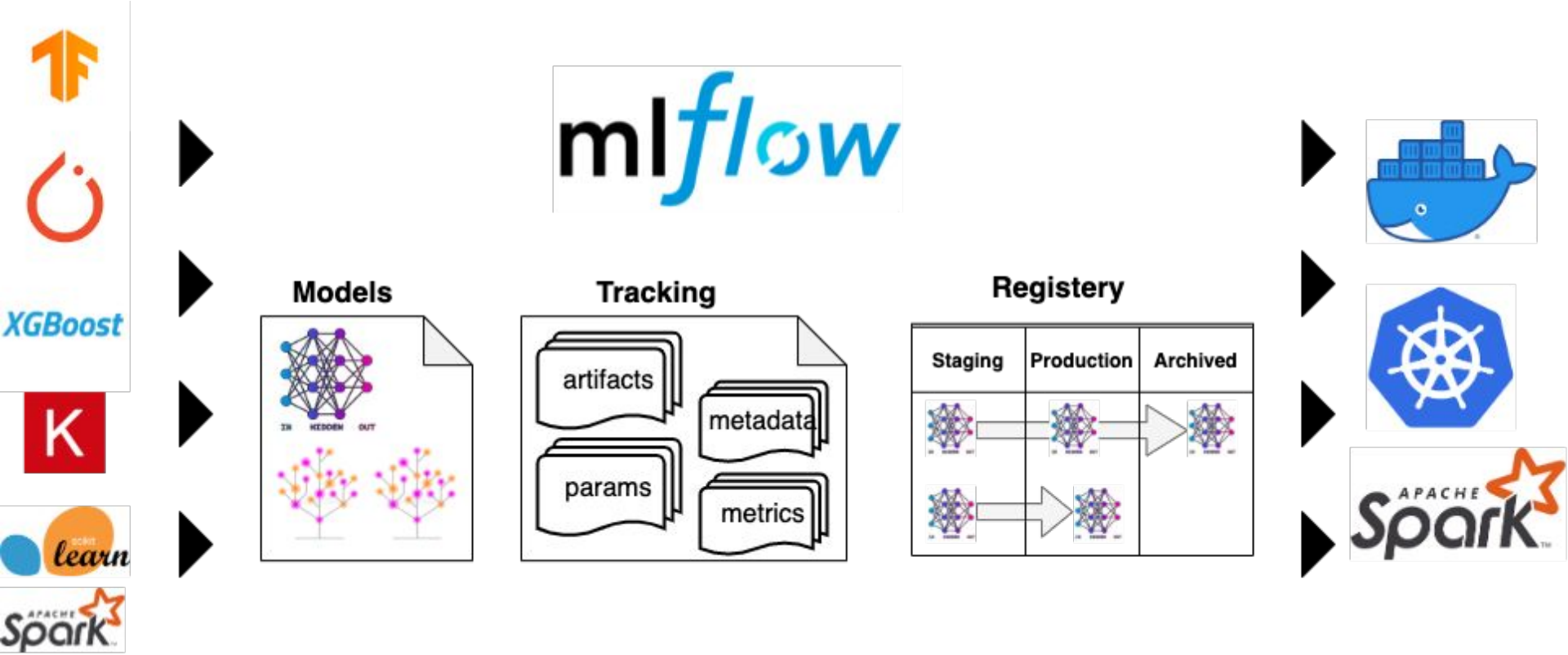
- Structured experiment tracking
- Centralized model registry
- Reproducible training pipelines
- Faster deployment cycle

MLflow is an end-to-end MLOps platform

It helps MARS to:

- Track experiments (parameters + metrics)
- Store and version models
- Compare different training runs
- Manage model lifecycle (staging → production)


1. Localhost (default)
Your ML code
mlflow Tracking APIs
Artifacts
Metadata
Local File
2. Localhost w/ various data stores
Your ML code
mlflow Tracking APIs
Artifacts
Metadata
Local File
Database
3. Remote Tracking w/ Tracking Server
Your ML code
Localhost
mlflow Tracking APIs
Artifacts
Metadata
mlflow Tracking Server
Cloud Storage
Database
Team


XGBoost
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learn
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mlflow
Models
IN
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OUT
Tracking
artifacts
metadata
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metrics
Registery
Staging
Production
Archived
APACHE
Spark

SageMaker Studio > Models > Registered Models > Iris Random Forest Model 37705e > Versions > Version 10 > Overview

Provide feedback

Applications (6)

JupyterLab | RStudio | Canvas | Code Editor | Studio Cl... | MLflow

Home

Running instances

Data

Auto ML

Experiments

Jobs

Pipelines

Version 10 Model Version

Overview | Activity | Details

Train	Evaluate	Audit	Deploy
Complete	Undefined	Draft	Pending Approval

Performance

Artifacts

Hyperparameters

Environment

Details

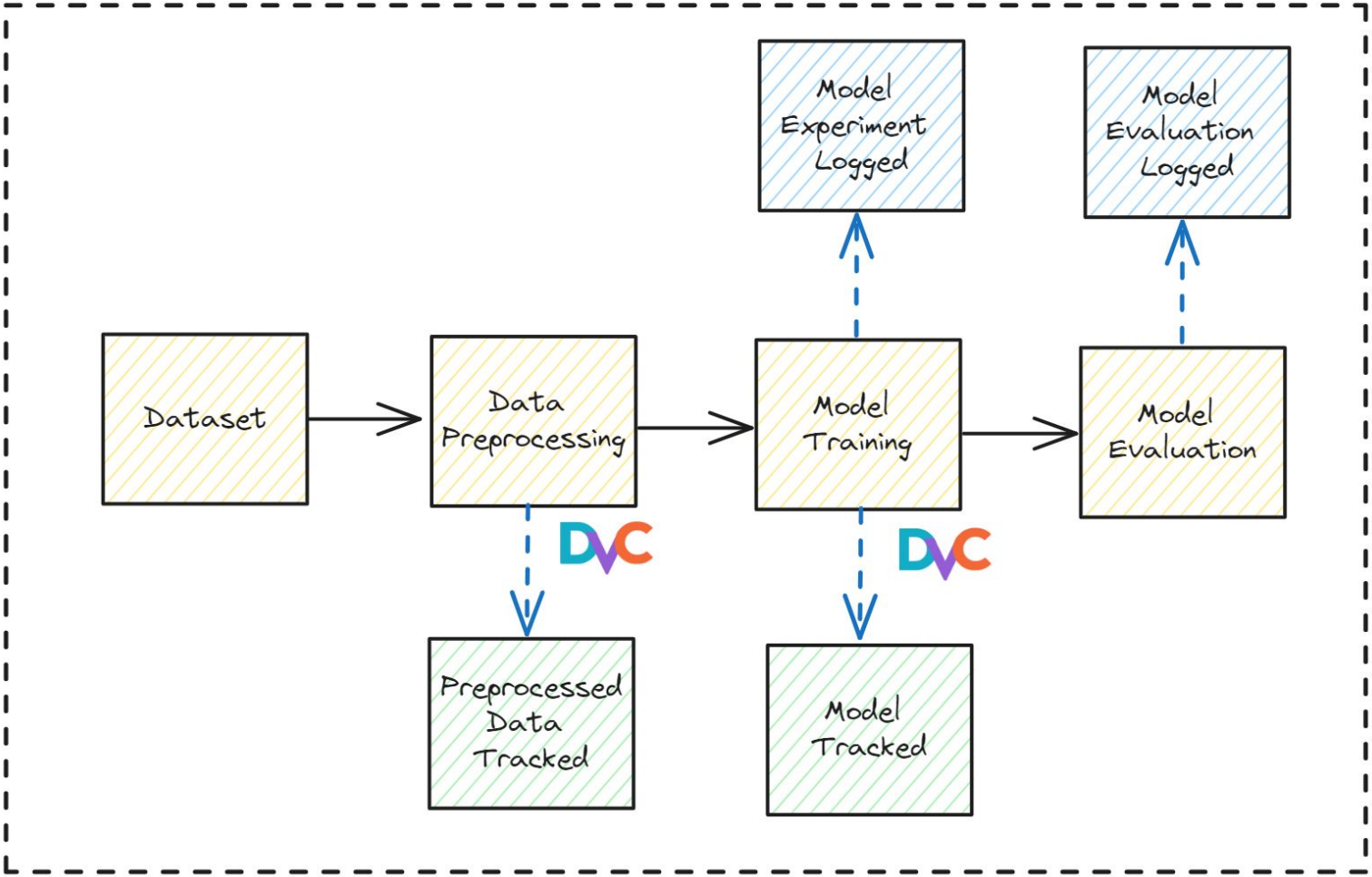
Metrics

Search...

Name	Value	Notes
accuracy	0.9555555555555556	--
precision	0.9573302469135803	--
recall	0.9555555555555556	--
f1_score	0.9557368557368557	--

4 results | Metrics per page 10 | Go to page 1 | Page 1 of 1

Experiment Tracking with MLFlow and DVC



Home – DVC

dvc.org

Guest (2)

Learn from AI, ML & data leaders from Dell, Lockheed Martin, Red Hat & more
AI-Ready Data Summit
On Demand WATCH NOW

DVC by lakeFS

Data Version Control | Doc | Blog | Course | Community | Get Started

Data Version Control

Manage data the way code is managed. Using a Git-like model we bring software engineering best practices to data, AI/ML and data science teams

What's new?

Data Version Control for AI/ML and Data Infrastructure

Get started with lakeFS — Free and open source

Book a Demo — lakeFS Enterprise

Data Version Control Git Extension for Small Data Science Projects

Get started with DVC — Free and open source

DVC for VS Code — Get VS Code extension

How to Control Data Version

	Open Source	Data Format Agnostic	Cloud/Storage Agnostic* (Supports most common cloud and storage types)	Simple to Use	Easy Support for BIG Data
DVC	✅ (Apache 2.0)	✅	✅	✅	❌
DELTA LAKE	✅ (Apache 2.0)	❌	✅	❌	✅
Git Large File Storage	✅ (MIT)	✅	❌	✅	❌
Pachyderm	✅⚠️ (Non-standard license)	✅	✅	❌	✅
DOLT	✅ (Apache 2.0)	❌	❌	❌	❌
lakeFS	✅ (Apache 2.0)	✅	❌	❌	✅

DAGsHub

MLflow Core Components

1. Tracking

- Logs parameters, metrics, images

2. Projects

- Reproducible ML code execution

3. Models

- Save and load trained models

4. Model Registry

- Version control for production models

Example pipeline:

- Input: Traffic camera images
- Model: YOLOv8 / Faster R-CNN
- Output: Car / Truck / Motorcycle detection

Metrics tracked:

- mAP (mean Average Precision)
- Precision / Recall
- Inference speed (FPS)

```
Dataset (Traffic Images)
        ↓
YOLOv8 Training (PyTorch)
        ↓
MLflow Tracking
- hyperparameters (lr, batch size)
- metrics (mAP, precision, recall)
        ↓
Model Comparison
        ↓
Best Model Selection
        ↓
MLflow Model Registry
        ↓
Deployment (API / Edge Device)
```

Multiple runs comparison

Run	LR	mAP50	Precision	Recall
A	0.01	0.68	0.74	0.65
B	0.001	0.79	0.85	0.77
C	0.0005	0.76	0.82	0.73

👉 MLflow automatically visualizes and compares runs

MLflow Model Registry at MARS

- Version control for AI models
- Staging → Production workflow
- Rollback capability
- Team approval system

Example:

- YOLOv8_v12 → Staging
- YOLOv8_v11 → Production
- YOLOv8_v10 → Archived

Business Impact

- ⏱️ 50% faster experiment tracking
- 📈 Better model accuracy selection
- 🔁 Full reproducibility of training
- 🧩 Centralized AI development system
- 🏭 Production-ready MLOps pipeline

Comparison for MARS use case

Feature	MLflow	Weights & Biases
Experiment tracking	Good	⭐ Excellent
Visualization	Basic UI	⭐ Very advanced
Model registry	⭐ Strong	Limited
Production MLOps	⭐ Strong	Medium
Hyperparameter tuning	Manual / custom	⭐ Built-in sweeps
Collaboration	Medium	⭐ Excellent
Deployment readiness	⭐ Strong	Limited

Best approach

Use MLflow when:

- Deploying models to production
- Managing model versions
- Building MLOps pipeline

Use W&B when:

- Doing research / experimentation
- Tuning YOLO models
- Visualizing training results

🚀 Final Recommendation

> For MARS:

- **MLflow = Production backbone**
- **W&B = Research + visualization tool**

👉 Best architecture: **Hybrid system (MLflow + W&B)**